PUNE INSTITUTE OF COMPUTER TECHNOLOGY, PUNE - 411043
Department of Computer Engineering
S.No.-27, Pune Satara Road, Dhankawadi, Pune-411043

PROJECT SYNOPSIS FORMAT

Academic Year -2025-26

Course Name & Code: Software Engineering

Project Title:

Machine Learning-Based Creditworthiness Prediction and Subsidy Matching System for Farmers Using Agricultural and Socio-Economic Data

Guide Name: Prof. Shweta Shah

1. Batch: E4

2. Date: 25/02/2026

3. Group Members

Roll No	Name
21407	Jyotiraditya Jadhav
21417	Shriya Mokashi
21418	Aryan Mutyal
21421	Ojas Sangwai

4.Title of the Project:

Machine Learning Based Creditworthiness Prediction and Subsidy Matching System for Farmers Using Alternative Agricultural and Socio-Economic Data

5. Introduction:

Agriculture remains a critical pillar of the economy in countries like India, yet many farmers face challenges in accessing formal credit due to lack of documented financial history and structured risk assessment mechanisms. Traditional credit evaluation models often fail to consider agriculture- specific risks such as climate variability,crop dependency , and market fluctuations.

This project proposes an AI-driven Farmer Credit Scoring and Subsidy Matching System that evaluates a farmer's creditworthiness using agricultural, financial, and risk- related parameters. By leveraging machine learning techniques, the system generates a standardized credit score and risk classification to assist financial institutions in making informed lending decisions.

Additionally, the system integrates policy support mapping to recommend suitable government subsidy schemes based on farmer profiles. The platform aims to promote financial inclusion , improve transparency in loan approvals, and support sustainable agricultural growth through data-driven decision-making.

6. Motivation:

Access to institutional agricultural credit remains a critical bottleneck in the rural financial ecosystem of India. A significant proportion of small and marginal farmers continue to rely on informal credit sources due to structural limitations in conventional banking risk assessment frameworks.

Despite the implementation of large-scale financial inclusion and agricultural risk mitigation initiatives such as Pradhan Mantri Fasal Bima Yojana and Pradhan Mantri Jan Dhan Yojana, formal credit penetration in rural agricultural segments remains suboptimal.

7. Problem Statement:

7.1 Existing System - The current agricultural lending framework relies on conventional credit scoring models designed primarily for salaried and formally employed urban populations. These systems assess creditworthiness using parameters such as:

- Credit bureau records (e.g., CIBIL score)
- Stable monthly income documentation
- Formal banking transaction history
- Collateral-backed lending
- Employment verification

Such models assume predictable and documented income streams. However, agricultural income is typically seasonal, weather-dependent, market-driven, and volatile, making it incompatible with standard financial risk assessment mechanisms.

7.2 Limitations of the Existing System

Due to the structural mismatch between traditional credit models and agricultural income patterns, farmers often lack:

- Formal credit history
- Documented and consistent income records
- Recognized collateral assets
- Digitized financial documentation

Consequently, financial institutions face difficulty in accurately evaluating farmer credit risk, resulting in high loan rejection rates, elevated perceived default risk, limited access to institutional credit, and dependence on informal lenders.

From a technical perspective, existing credit systems do not incorporate agriculture-specific risk indicators into their evaluation framework. Critical parameters that remain unaccounted for include:

- Crop productivity and yield stability
- Landholding size and verified ownership
- Irrigation infrastructure availability
- Participation in government insurance or subsidy schemes
- Historical agricultural performance trends
- Crop diversification and market linkage
- Regional agro-climatic risk factors

7.3 Need for the Proposed System

There is a need for a farmer-centric credit assessment framework that integrates agricultural, environmental, and socio-economic indicators into a unified predictive model. The proposed **Machine Learning-Based Creditworthiness Prediction and Subsidy Matching System** will:

- Utilize alternative agricultural and socio-economic data

- Apply supervised learning techniques to estimate probability of default
- Generate a normalized Farmer Credit Risk Score
- Provide explainable outputs for transparency
- Align risk evaluation with agricultural income dynamics

The system aims to establish a scalable, data-driven, agriculture-specific credit evaluation mechanism that enhances financial inclusion and enables informed lending decisions in rural ecosystems.

8. Objectives:

8.1 Primary Objective

To design and implement a Machine Learning-Based Creditworthiness Prediction and Subsidy Matching System for Farmers that evaluates credit risk using alternative agricultural and socio-economic data instead of traditional credit bureau–based models.

8.2 Specific Objectives

A. Credit Risk Prediction

1. Develop a supervised ML model to estimate probability of loan default.
2. Generate a normalized Farmer Credit Risk Score (0–100).
3. Classify farmers into risk categories (Low, Medium, High).
4. Incorporate agriculture-specific variables such as land size, yield, irrigation, and loan history.

B. Alternative Data Integration

5. Utilize agricultural productivity and regional indicators for contextual risk assessment.
6. Design a feature engineering pipeline aligned with the training dataset.

C. Subsidy Matching

7. Implement eligibility-based subsidy matching using farmer profile data.
8. Provide scheme recommendations based on predefined criteria.

D. Transparency and Scalability

9. Provide explainable outputs indicating key factors influencing the score.
10. Ensure modular, scalable architecture for integration with financial institutions.

9. Scope of the Project:

9.1 In Scope: Features and functionalities included

- Farmer registration and authentication
- Collection of financial, agricultural and location based inputs
- Document upload and authenticity verification
- Integration of external climate , market and policy datasets
- Feature engineering for financial and risk indicators
- Machine-learning based probability of default estimation
- Credit score generation (0-100 scale)
- Risk categorization (low/medium/high)
- Subsidy recommendation based on farmer profile and policy mapping

9.2 Out of scope : What is intentionally excluded

- Direct loan disbursement or financial transactions
- Full integration with banking core systems
- Real - time integration with national credit bureaus
- Automatic government subsidy approval processing
- Certified financial or legal advisory services
- National- level fraud detection systems beyond document upload verification

10. Software and Hardware Requirements:

10.1 Software Requirements (client and developer side)

A. Client Side Requirements

For Farmers / Field Agents (Mobile Users):

- Web Browser: Google Chrome (latest stable version)
- Internet Connectivity: 4G / 5G
- SMS Capability (for OTP/notifications)
- PDF Viewer (for downloadable reports)

For Admin / Analyst (Desktop Users):

- Operating System: Windows 10 / Windows 11 / macOS 12+ / Ubuntu 20.04+
- Web Browser: Google Chrome (latest stable version)
- PDF Reader
- Spreadsheet software (MS Excel or equivalent) for report analysis

B. Developer Side Requirements

Frontend Development

- React v18
- npm v9+
- Visual Studio Code (latest version)
- Git v2.30+

Backend Development

- Python v3.1
- Node.js
- FastAPI (latest stable version)
- Uvicorn (latest stable version)
- PostgreSQL v14

Machine Learning Development

- Python v3.11
- pandas v1.5+
- numpy v1.23+
- scikit-learn v1.2+
- XGBoost v1.7+
- SHAP (latest stable version)
- MLFlow
- Jupyter Notebook (latest stable version)

10.2 Hardware Requirements (client and developer Side)

A. Client Side Hardware

Farmer / Field Agent Device

- Android Smartphone
- Minimum 4 GB RAM
- Minimum 64 GB internal storage
- Camera (for document uploads)
- GPS capability

Admin / Analyst System

- Laptop or Desktop Computer
- Intel i5 / Ryzen 5 processor or equivalent

- 16 GB RAM
- 512 GB SSD
- 1080p Display
- Stable broadband internet connection

B. Developer Side Hardware

Development Machine

- Intel i7 / Ryzen 7 processor or equivalent
- 16–32 GB RAM
- 512 GB SSD or higher
- Dedicated GPU (NVIDIA RTX series, optional for ML training)
- Stable high-speed internet connection

Server (Production Deployment)

- 4–8 vCPU
- 16–32 GB RAM
- 200 GB SSD storage
- Linux-based OS (Ubuntu 22.04 LTS)
- Dedicated cloud hosting with HTTPS enabled
- Daily automated backup support

11. Proposed System / Methodology:

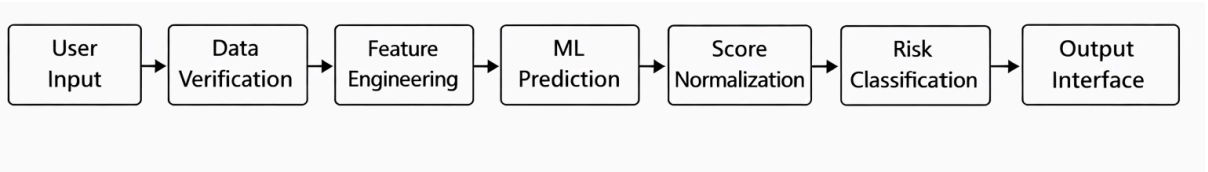
11.1 System Overview

The Credit Scoring Module is designed as a data-driven predictive system that evaluates farmer creditworthiness using alternative agricultural and socio-economic indicators.The module performs the following sequential operations:

1. Data acquisition through structured web-based input forms
2. Multi-layer data validation and authenticity verification
3. Feature extraction and transformation
4. Machine Learning-based probability estimation
5. Credit score generation and risk classification
6. Output presentation with explainable insights

The system outputs a normalized **Farmer Credit Risk Score (0–100)** along with a corresponding risk category and decision-support recommendations.

11.2 System Architecture – Credit Module Workflow



Step 1: Data Acquisition Layer

The system collects structured data from two sources: farmer inputs through the web interface and external APIs. The collected data corresponds to the variables used in the AgriRiskFin dataset.

A. Farmer-Provided Inputs

Agricultural & Enterprise Details:

- Enterprise size (Small / Medium / Large)
- District / State (mapped to Region)
- Crop type
- Landholding size
- Irrigation type

Financial Details:

- Annual revenue
- Annual expenses
- Existing loan amount
- Approximate asset value
- Previous loan default history

B. Automatically Fetched Data (Mapped to Dataset Variables)

Based on location and crop type, the system retrieves:

- Average temperature
- Rainfall
- Drought index
- Flood risk score
- Commodity price index
- Input cost index
- Policy support score

These correspond directly to the climate, market, and policy variables in the training dataset.

Step 2: Data Verification and Authenticity Layer

To ensure reliability and prevent fraudulent entries, the system integrates document-based and logical verification mechanisms.

A. Document Upload Verification

The system supports secure upload and validation of:

- Land ownership proof (7/12 extract)
- Aadhaar (masked storage)
- Crop insurance certificate
- Bank passbook snapshot

B. External Dataset Cross-Validation

Where available, the system cross-verifies inputs with:

- Government district-level agricultural yield datasets
- District-level rainfall or agro-climatic datasets

C. Logical Consistency Checks

The system applies rule-based validation to detect anomalies, such as:

- Extremely high yield relative to district averages
- Income inconsistent with land size or crop type
- Irrigation claim inconsistent with regional infrastructure data

Flagged records are either rejected or routed for manual review.

Step 3: Feature Engineering Layer

The Feature Engineering Layer transforms validated raw inputs into structured, model-ready features consistent with the AgriRiskFin training dataset. This layer ensures that real-time farmer data aligns with the statistical structure and feature space used during model training.

It performs ratio computation, risk indexing, encoding, normalization, and feature vector construction before passing data to the Machine Learning Prediction Layer.

3.1 Financial Feature Construction

From farmer-provided financial inputs, the system derives standardized financial stability indicators:

- Expense-to-Revenue Ratio (ERR)
  ERR = Expenses/Revenue
- Debt Sustainability Ratio (DSR)
  DSR = Loan Amount/Revenue
- Net Profit Margin (NPM)
  NPM = Net Profit/Revenue
- Debt-to-Equity Ratio (D/E)
  Computed using loan amount and declared asset value.

These indicators measure repayment burden, operational efficiency, leverage, and profitability which are core predictors of financial risk.

3.2 Climate Risk Feature Construction

Using externally fetched climate data mapped to the farmer's location, the system constructs environmental risk indicators:

- Climate Resilience Index (CRI)
  Computed using normalized Rainfall, Drought Index, and Flood Risk Score.
- Temperature Stress Indicator
  Derived from average temperature thresholds relevant to crop sustainability.

These features capture environmental volatility that directly impacts agricultural income stability.

3.3 Market Risk Feature Construction

From market data:

- Market Risk Indicator (MRI)
  MRI = Commodity Price Index/Input Cost Index

This reflects margin pressure due to output price fluctuation and rising input costs.

3.4 Policy and Regional Encoding

- Farmer district/state is mapped to the dataset-level Region variable.
- Policy Support Score is included as a model feature.
- Enterprise Size is encoded for model compatibility.

3.5 Data Transformation and Alignment

Before model invocation, the system performs:

- Numerical feature normalization/standardization

- Categorical encoding (e.g., One-Hot Encoding)
- Missing value handling
- Outlier range validation
- Construction of final structured feature vector

The output of this layer is a normalized feature vector strictly aligned with the training dataset schema and model input format.

3.6 Output

The final processed feature vector is forwarded to the Machine Learning Prediction Layer for financial risk classification and credit score generation.

Step 4: Machine Learning Prediction Layer

The system employs supervised classification algorithms trained on historical labeled data (default / non-default).

Candidate Algorithms:

- Logistic Regression
- Random Forest Classifier
- Gradient Boosting Classifier

The trained model outputs:

$$P(\text{default}) \in [0,1]$$

This represents the estimated probability that the farmer may default on a loan.

Step 5: Credit Score Computation

The probability of default is transformed into a normalized credit score:

$$\text{Credit Score} = (1 - P(\text{default})) * 100$$

This produces a score between 0 and 100, where:

- Higher score → Lower credit risk
- Lower score → Higher credit risk

Step 6: Risk Categorization

The generated score is mapped to predefined risk categories:

Credit Score Range	Risk Category
80 – 100	Low Risk
50 – 79	Medium Risk
0 – 49	High Risk

The final output includes:

- Credit Score
- Risk Category
- Key contributing factors (Explainable AI output)
- Basic lending recommendation

11.3 Design Characteristics

The proposed credit module is:

- Agriculture-specific
- Data-driven
- Modular and scalable
- Explainable
- Compatible with integration into lender dashboards

12. Theory / System Description:

Credit Scoring Logic

12.1 Conceptual Foundation

The Credit Scoring Module is grounded in the theory of probability-based credit risk modeling, where a borrower's creditworthiness is expressed as the likelihood of default under uncertain financial and environmental conditions.

Traditional credit systems rely primarily on collateral and past repayment history. However, agricultural borrowers are exposed to additional systemic risks such as climate variability, commodity price volatility, and seasonal income fluctuations. Therefore, the proposed model

incorporates multidimensional agricultural risk indicators to estimate repayment capacity more accurately.

The theoretical objective of the module is to estimate:

$$P(\text{Default} \mid \text{Financial, Climate, Market, Policy Factors})$$

12.2 Risk Estimation Principle

The system treats credit risk as a binary classification problem, where:

- Class 0 → Non-default (financially stable)
- Class 1 → Default (financially risky)

Using supervised learning, the model identifies nonlinear relationships between agricultural variables and financial distress patterns observed in the training dataset. Instead of relying on a single factor (e.g., income), the model evaluates combined risk exposure through:

- Financial leverage burden
- Income stability
- Environmental vulnerability
- Market pressure
- Institutional policy support

The model output is a probability value between 0 and 1, representing estimated default likelihood.

12.3 Credit Score Transformation Logic

The predicted probability of default is converted into a normalized credit score on a 0–100 scale.

The transformation follows an inverse-risk principle:

- Higher default probability → Lower credit score
- Lower default probability → Higher credit score

This ensures interpretability and consistency with conventional credit scoring systems. The normalization enables:

- Standardized comparison across farmers
- Clear threshold-based decision support
- Integration with subsidy and lending modules

12.4 Explainability and Transparency

To promote transparency and user trust, the system supports explainable outputs. The model highlights the most influential contributing factors affecting the predicted risk outcome.

This allows:

- Farmers to understand improvement areas
- Lenders to justify decisions
- Policymakers to analyze systemic vulnerabilities

The scoring logic therefore balances predictive performance with interpretability.

Subsidy Matcher Module:

The Subsidy Matcher Module is designed to identify and recommend relevant government agricultural subsidy schemes based on the farmer's profile,financial condition,crop details and regional parameters.The module follows a rule-based and profile matching approach to determine eligibility.

12.5 Purpose:

The purpose of the Subsidy Matcher Module is to improve awareness and accessibility of government schemes by automatically mapping farmer data with eligibility criteria of various subsidy programs

1. Input Parameters

   The module uses the following inputs :-

   - Enterprise size(small/medium/large)
   - Landholding Size
   - Crop type
   - Region/District
   - Irrigation type
   - Revenue and income details
   - Policy support Score(if derived)

2. Matching Logic

   The module applies predefined eligibility rules stored in the system database.

   - The system checks landholding criteria
   - Validates crop-specific eligibility
   - Verifies regional applicability

- Checks income threshold conditions
- Evaluates irrigation or infrastructure eligibility

If all the required conditions are marked as satisfied , the scheme is marked as eligible. The system may also assign a matching score based on the number of satisfied conditions.

3. Processing Flow

    - Farmer submits profile data
    - System retrieves subsidy databases
    - Eligibility rules are evaluated
    - Matching schemes are filtered
    - Eligible schemes are displayed on the dashboard

4. Output(what the model will display)

    - List of eligible schemes
    - Brief description
    - Eligibility reasoning
    - Basic benefits information

5. Design characteristics

    The subsidy Matcher Module is :

    - Rule based and deterministic
    - Modular and scalable
    - Updated with new government schemes
    - Independent from credit score computation

13. Functional Requirements:

Credit Module:

1. Risk Probability Estimation - The system shall compute the probability of default using the trained machine learning model.
2. Credit Score Normalization - The system shall transform the predicted probability into a standardized credit score ranging from 0 to 100.
3. Risk Band Classification - The system shall categorize the generated credit score into predefined risk bands (Low, Medium, High).
4. Explainable Output Generation - The system shall provide key contributing risk indicators influencing the final score.

5. Decision Support Output - The system shall generate advisory-level lending recommendations based on the assigned risk band.
6. Integration Compatibility - The credit module shall provide structured output that can be integrated with subsidy matching and lender dashboards.

UI + Dashboard (Functional Requirements) :-

User registration and login

- The system shall allow farmers to register using mobile number/email
- The system shall provide secure login authentication
- The system shall validate user credentials before getting access

Data Entry Interface

- The system shall provide structured forms for enabling financing and agricultural details
- The system shall validate mandatory fields before submission
- The system shall restrict invalid numerical inputs

Document Upload Module

- The system shall allow users to upload verification documents
- The system shall store uploaded documents securely

Credit Sore Dashboard

- The system shall display a normalized credit score (0-100)
- The system shall display the risk category (Low/Medium/High)

Navigation and stability

- The system shall provide a responsive interface(mobile + desktop)
- The system shall allow users to edit submitted data before final submission

Subsidy Recommendation Dashboard

- The system shall display eligible government schemes
- The system shall provide basic scheme details

14. Non-Functional Requirements:

14.1 Performance

Performance refers to the system's ability to process inputs and generate outputs within acceptable time and resource constraints.

The system follows principles of computational efficiency and bounded latency. Since the credit scoring module involves real-time model inference, performance optimization is essential to ensure responsiveness under realistic usage conditions.

The system shall:

- Maintain bounded response latency for prediction requests under normal operational load.
- Optimize inference complexity by deploying pre-trained models and minimizing runtime feature computation overhead.
- Employ efficient data pipelines to reduce I/O bottlenecks.
- Support concurrent multi-user access using asynchronous request handling mechanisms.

14.2 Security

Security addresses protection against unauthorized access, data breaches, and malicious manipulation. The system processes personally identifiable information (PII), financial indicators, and land-related records; therefore, it must adhere to secure system design principles.

The system shall:

- Encrypt all communication channels using secure transport-layer protocols.
- Implement encrypted storage for sensitive user documents and identifiers.
- Utilize token-based authentication mechanisms to prevent session hijacking.
- Implement role-based access control (RBAC) to restrict system capabilities based on user roles.
- Protect against common web vulnerabilities such as injection attacks and cross-site scripting.

14.3 Scalability

Scalability refers to the system's ability to handle growth in users, data volume, and computational demand without architectural redesign. The system adopts modular and

distributed architecture principles to support horizontal scalability. Rather than vertically increasing hardware capacity, the system allows service replication and load distribution.

The system shall:

- Support containerized deployment for independent service scaling.
- Allow independent scaling of the ML inference service and API service.
- Maintain database performance under increasing record volume through indexing and optimized schema design.
- Support stateless API design to enable load-balanced distributed deployments.

14.4 Usability

Usability concerns the human-system interaction dimension, ensuring accessibility, clarity, and ease of use across varying literacy and technological familiarity levels. The system design follows user-centered design principles and accessibility standards.

The system shall:

- Provide mobile-responsive interface design suitable for rural smartphone usage.
- Include structured form validation to prevent incorrect data entry.
- Provide clear visual feedback and error messages.
- Support multilingual interface capability for regional adoption.
- Present outputs in simplified, interpretable dashboards rather than raw probability values.

14.5 Maintainability

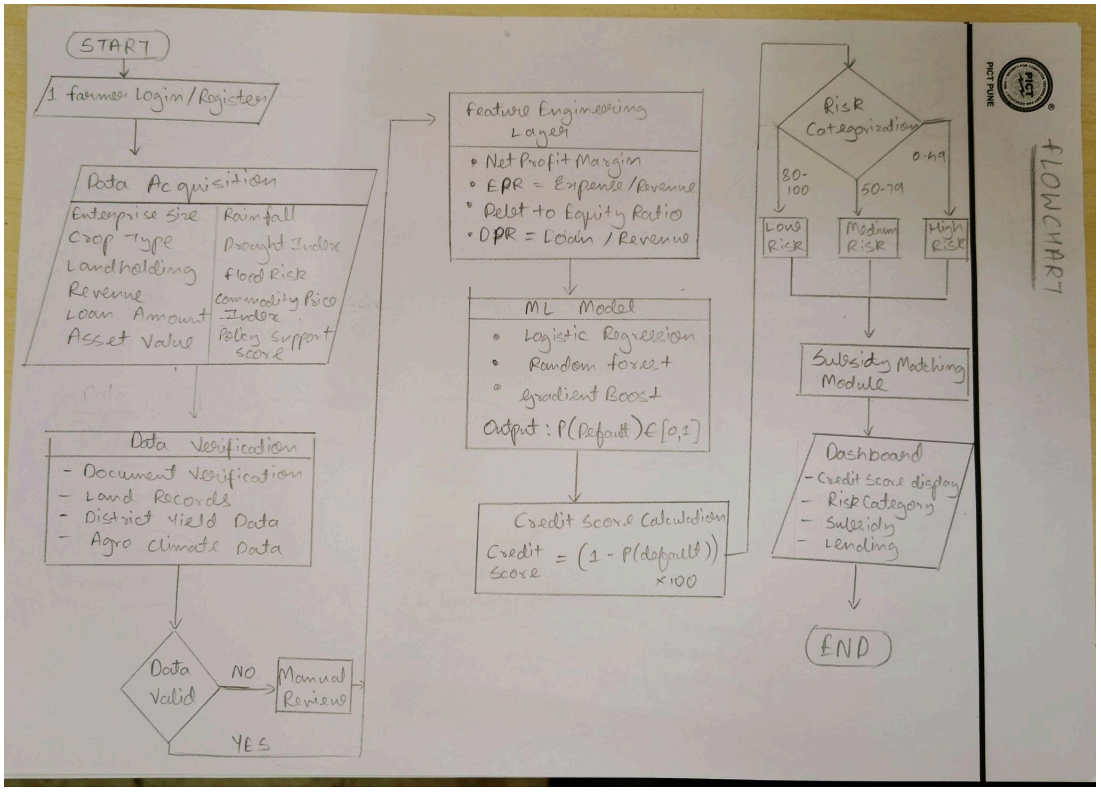
Maintainability refers to the ease with which the system can be modified, extended, debugged, and improved. Given the iterative nature of machine learning systems, maintainability is critical for retraining, updating policy rules, and refining risk thresholds.

The system shall:

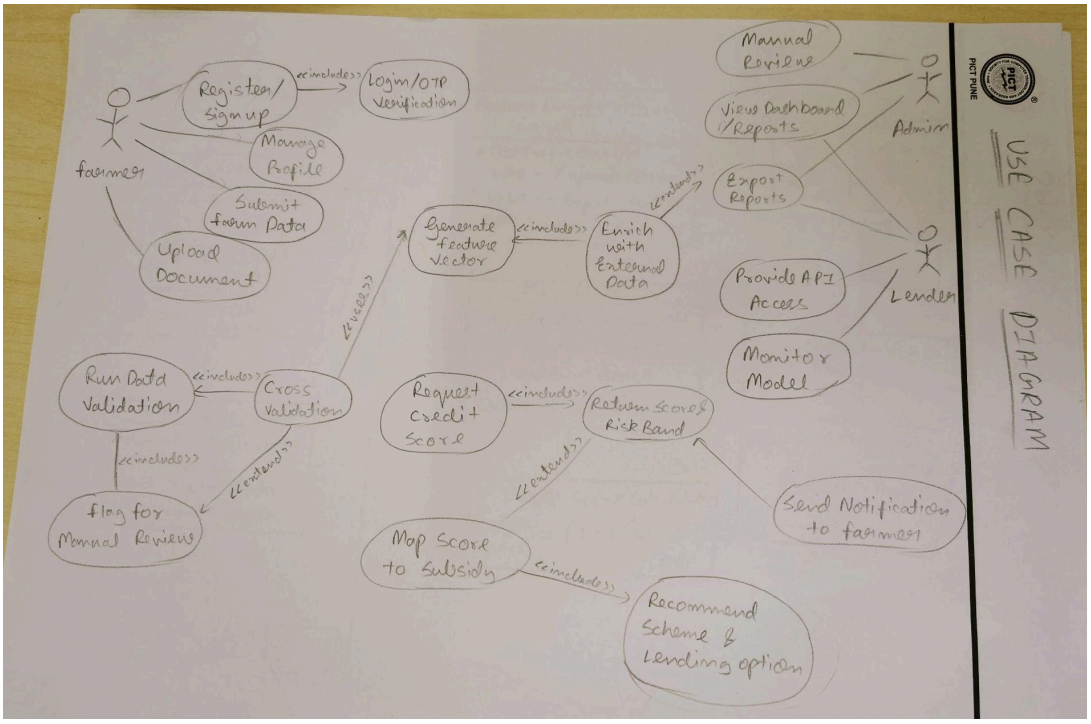
- Follow modular architecture separating frontend, backend, and ML components.
- Maintain clear API contracts and documentation.
- Use version control systems to track model and code evolution.
- Enable CI/CD pipelines for structured testing and deployment.
- Support periodic model retraining without disrupting system services.

15. System Design:

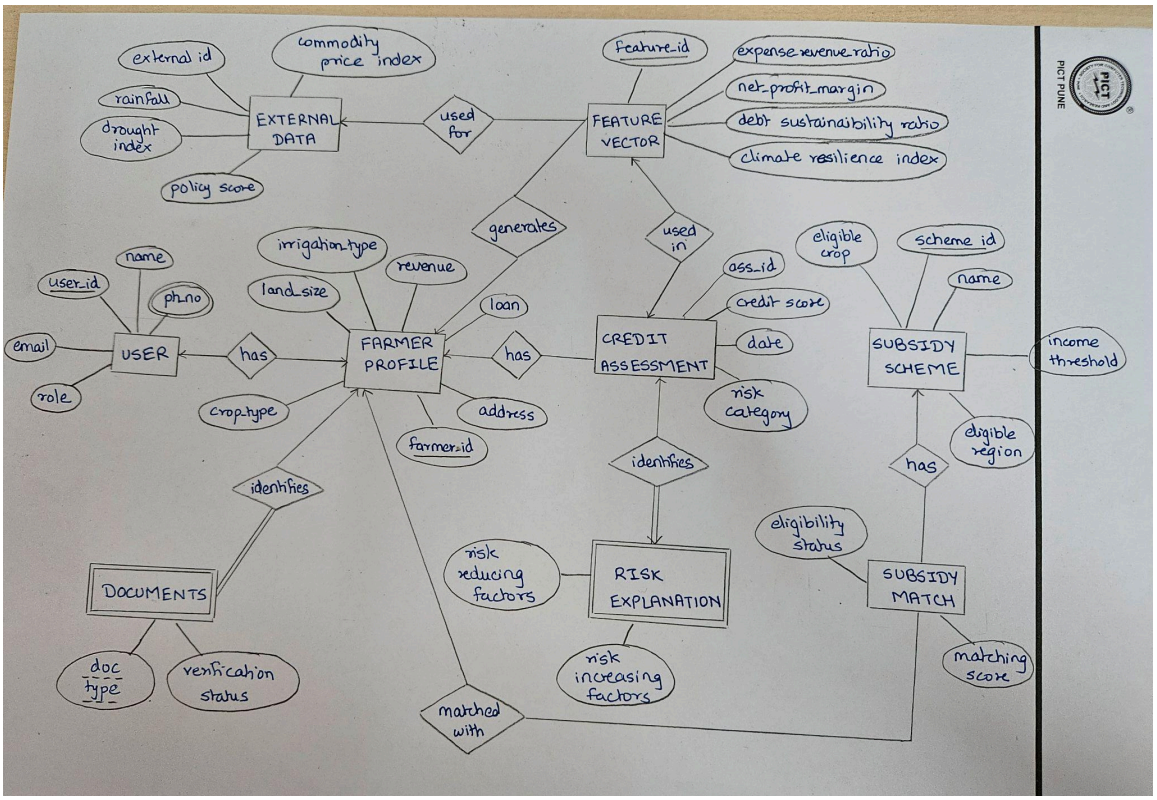
15.1 Flowchart



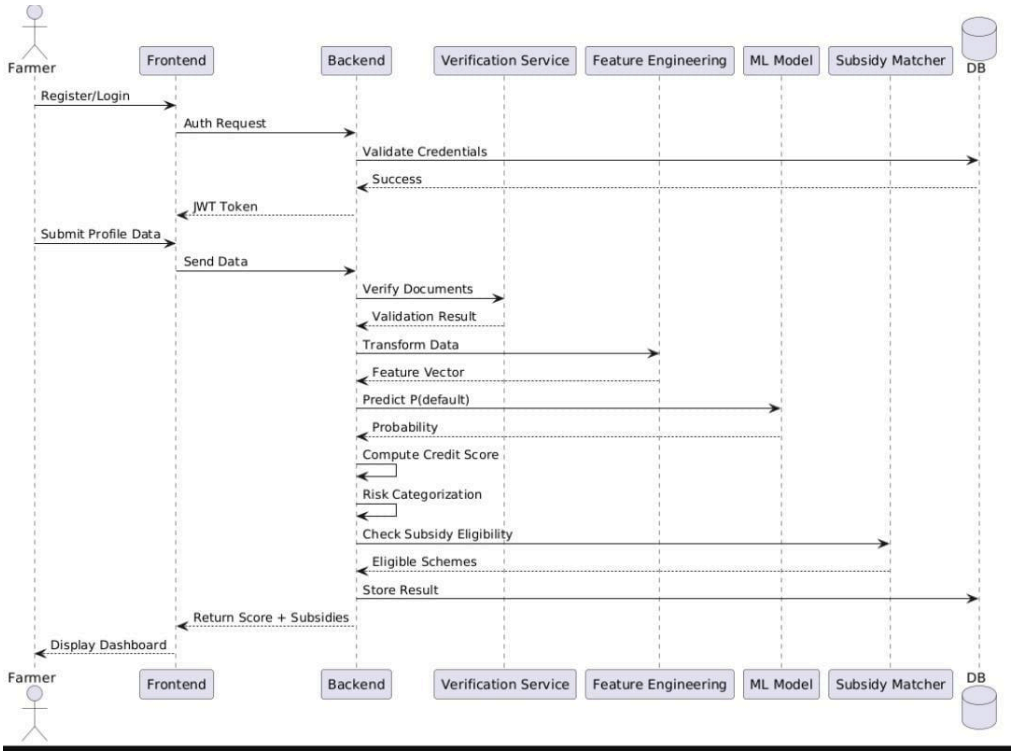
15.2 Use Case Diagram



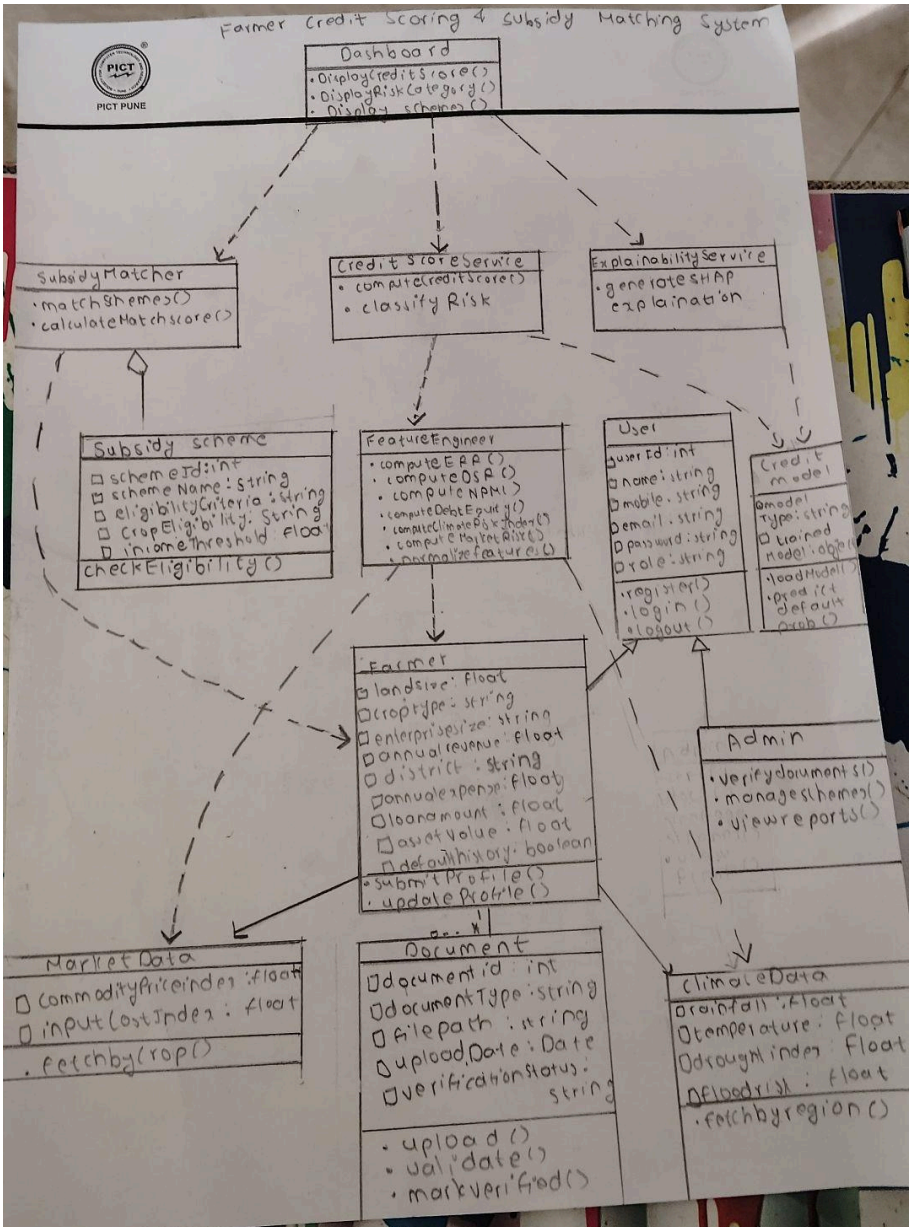
15.3 ER Diagram



15.4 Sequence Diagram



15.5 Class Diagram



16. Project Plan & Timeline:

Phase	Activities	Duration(require for the Task)

1	Problem analysis & Requirement Gathering	16 Jan - 23 Jan (1 week)
2	System design (UI design + architecture + feature planning)	23 Jan - 6 Feb (2 weeks)
3	Implementation (Frontend + Backend + Ml Development)	6 Feb - 7 March (3/4 weeks)
4	Testing (Unit Testing , Integration Testing, UI testing)	7 March - 21 March (2 weeks)
5	Documentation & Final Review	21 March - 1 April (2 weeks)

17. Risks & Challenges:

Risk	**Details**
Data Availability & Quality	Limited labeled repayment data and inconsistent agricultural inputs may affect prediction accuracy. **Mitigation:** Use proxy labels, integrate verified external datasets, apply strong validation rules, and conduct pilot data collection.
Label Scarcity	Lack of reliable default history may reduce supervised learning effectiveness. **Mitigation:** Use expert-labeled samples, semi-supervised techniques, and ensemble learning methods.

Privacy & Regulatory Compliance	Handling personal and land-related data may create legal and data protection concerns. **Mitigation:** Obtain explicit consent, mask identifiers, encrypt data, and comply with data protection standards.
Model Bias & Fairness	Model predictions may unintentionally disadvantage certain farmer groups. **Mitigation:** Conduct fairness evaluation, monitor subgroup performance, and allow human review for sensitive decisions.
Usability & Adoption	Farmers may face difficulty using the digital platform due to language or technical barriers. **Mitigation:** Provide local-language interface, simplified UI, and field-agent assistance.
Time Constraints	Limited academic timeline may restrict full-scale testing and validation. **Mitigation:** Prioritize MVP features and follow structured milestone planning.

18. Expected Outcomes & Deliverables:

18.1 Expected Outcomes

The proposed Farmer Credit Scoring and Subsidy Matching System is expected to achieve the following outcomes:

1. Accurate Credit Risk Assessment

The system will generate a normalized credit score (0–100) and classify farmers into Low, Medium, or High risk categories using a machine learning–based probability model.

2. Agriculture-Specific Risk Evaluation

The credit scoring mechanism will incorporate financial, climate, market, and policy-related factors to provide context-aware risk analysis tailored to agricultural enterprises.

3. Improved Financial Decision Support

The system will assist:

- Farmers in understanding their financial standing
- Lenders in making informed lending decisions
- Policymakers in identifying vulnerable farmer segments

4. Automated Subsidy Eligibility Identification

Based on credit score, region, and policy support indicators, the system will recommend suitable government schemes and subsidy programs.

5. Transparent and Explainable Results

The system will provide explainable outputs by highlighting key factors contributing to the predicted risk level.

18.2 Deliverables

The project will deliver the following tangible components:

1. Web-Based Application

A functional web platform with:

- Farmer registration and login
- Data input forms
- Document upload system
- Credit score display dashboard

2. Machine Learning Model

- Trained supervised classification model
- Feature-engineered dataset
- Model evaluation report (Accuracy, Precision, Recall, F1-score)
- Saved model file for deployment

3. Backend API System

- Secure REST API for prediction requests
- Feature transformation module
- Model inference endpoint
- Structured JSON output response

4. Dataset & Documentation

- Processed training dataset
- Feature engineering documentation
- Model training and testing report
- System design diagrams

5. Credit Score & Risk Framework

- Defined credit score computation logic
- Risk categorization thresholds
- Explainability framework

19. Ethical, Security & Legal Considerations:

19.1 Data privacy

The system processes personally identifiable information (PII), financial records, land ownership details, and socio-economic indicators. Therefore, it must adhere to foundational data protection principles derived from modern data governance frameworks.

A. Informed and Explicit Consent

Before any data collection, the system shall obtain explicit, informed, and unambiguous consent from the user.

From a legal and ethical standpoint, consent must satisfy:

- Specificity – Data must be collected for clearly defined purposes.
- Informed Understanding – Users must be aware of what data is collected and why.
- Voluntariness – Consent must not be coerced or implied.
- Revocability – Users must be able to withdraw consent.

B. Masking and Encryption of Sensitive Identifiers

Sensitive identifiers such as Aadhaar numbers, bank details, and land records shall:

- Be masked in user-visible dashboards.
- Be encrypted at rest using strong encryption standards (e.g., AES-256).
- Be transmitted using secure communication protocols (TLS 1.2+).

C. Regulatory Compliance

The system shall comply with:

- Applicable national data protection laws.
- Financial data handling guidelines.
- Public policy transparency requirements.

19.2 Bias and fairness

A. Subgroup Fairness Evaluation
The model shall be evaluated across different farmer segments such as region, landholding size, and crop type to ensure that predictive performance does not disproportionately disadvantage any particular group.

B. Bias Mitigation and Human Oversight
Where bias is detected, corrective measures such as reweighting, threshold adjustment, or sampling techniques shall be applied. Additionally, high-risk classifications shall be subject to human review to prevent automated unfair decisions.

19.3 Transparency:

The system shall provide interpretable explanations for each credit score generated. Given that machine learning models may involve nonlinear relationships and feature interactions, post-hoc interpretability techniques such as feature importance ranking or SHAP-based contribution analysis shall be employed. For each prediction, the dashboard shall display:

- Key risk-increasing factors
- Key risk-reducing factors
- Relative magnitude of influence

This transforms abstract probability outputs into actionable insights. For example, high debt-to-revenue ratio or climate volatility indicators may be highlighted as contributing to elevated risk. By making the reasoning behind the score transparent, the system reduces the opacity commonly associated with automated decision systems.

20. Software Engineering (SE) Model to Be Used:

20.1 Name of the SE Model

Agile Software Development Model (Iterative and Incremental Approach)

20.2 Why This Model is Suitable for the Project

The proposed Farmer Credit Scoring and Subsidy Matching System involves iterative model development, continuous testing, and progressive refinement of features. Since machine learning systems require repeated experimentation, validation, and tuning, a rigid linear model such as Waterfall would not be appropriate.

The Agile model is suitable because:

- It supports incremental development of modules (credit scoring, verification, subsidy mapping).
- It allows continuous improvement of the ML model based on new data and evaluation metrics.
- It enables parallel development of frontend, backend, and ML components.
- It provides flexibility to refine business rules (e.g., risk thresholds, subsidy logic).
- It supports early delivery of a Minimum Viable Product (MVP) for pilot testing.

Thus, Agile ensures adaptability, faster feedback cycles, and improved system reliability.

20.3 Key Phases Involved

The project follows iterative Agile sprints consisting of the following phases:

1. Requirement Analysis

- Problem identification
- Stakeholder requirement gathering
- Dataset understanding
- Functional and non-functional requirement specification

2. System Design

- Architecture design (Frontend, Backend, ML module)
- Database schema design
- API design
- Flowcharts and UML diagrams

3. Implementation

- Frontend development (React)
- Backend API development (FastAPI)
- Feature engineering and model training
- Integration of external APIs

4. Testing & Validation

- Unit testing of modules
- Model performance evaluation (Accuracy, Precision, Recall, F1-score)
- Fairness and bias checks
- Usability testing

5. Deployment & Monitoring

- Model deployment using containerization
- Cloud hosting setup
- Performance monitoring
- Periodic model retraining

21.Sustainability and Feasibility Analysis:

21.1 Sustainability Goals Identified

- SDG 1 – No Poverty
- SDG 2 – Zero Hunger
- SDG 8 – Decent Work and Economic Growth
- SDG 10 – Reduced Inequalities

Justification

- The proposed ML-Based Creditworthiness Prediction and Subsidy Matching System improves access to institutional credit for small and marginal farmers.
- Supports poverty reduction (SDG 1) by enabling fair access to formal financial services.
- Strengthens agricultural productivity and food security (SDG 2) through better financial access.
- Promotes rural economic growth (SDG 8) via data-driven lending decisions.
- Reduces financial inequality (SDG 10) by replacing urban-centric credit models with agriculture-specific risk assessment.

21.2 Technical Feasibility

- Uses supervised ML algorithms suitable for structured agricultural data.
- Modular architecture allows scalable web-based deployment.
- Integrates alternative agricultural and socio-economic indicators into predictive modeling.

21.3 Economic Feasibility

- Utilizes existing datasets and open-source ML libraries.
- Reduces operational costs for lenders through automated risk assessment.
- Requires minimal additional infrastructure.

21.4 Social and Environmental Sustainability

- Improves financial inclusion and reduces reliance on informal lenders.
- Enhances access to government subsidies and insurance schemes.

- Encourages informed agricultural financing aligned with regional productivity and risk factors.

22. References:

- Khandani, A. E., Kim, A. J., & Lo, A. W. (2010). Consumer credit-risk models via machine-learning algorithms. *Journal of Banking & Finance*, 34(11), 2767–2787. https://doi.org/10.1016/j.jbankfin.2010.06.001
- Dumitrescu, E., Hué, S., Hurlin, C., & Tokpavi, S. (2022). Machine learning for credit scoring: Improving logistic regression with non-linear decision-tree effects. *European Journal of Operational Research*, 297(3), 1178–1192. https://doi.org/10.1016/j.ejor.2021.06.053
- Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems (NeurIPS)*, 30, 4765–4774.
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794. https://doi.org/10.1145/2939672.2939785
- Aker, J. C., & Mbiti, I. M. (2010). Mobile phones and economic development in Africa. *Journal of Economic Perspectives*, 24(3), 207–232. https://doi.org/10.1257/jep.24.3.207
- Government of India, Ministry of Agriculture & Farmers Welfare. (2023). *Pradhan Mantri Fasal Bima Yojana (PMFBY) – Operational Guidelines*. Department of Agriculture and Farmers Welfare. https://pmfby.gov.in
- Reserve Bank of India. (2021). *Report of the Internal Working Group to Review Agricultural Credit*. RBI Publications. https://www.rbi.org.in
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32. https://doi.org/10.1023/A:1010933404324

23. Conclusion:

The proposed Farmer Credit Score Predictor and Subsidy Matcher presents a structured , data-driven approach to agricultural credit evaluation. By integrating financial indicators, climate risk factors , market dynamics, and policy support parameters , the system enhances transparency and fairness in credit assessment.

Through machine learning - based risk prediction and explainable outputs, the platform supports informed lending decisions while promoting financial inclusion among farmers. Additionally, the subsidy recommendation component strengthens access to government support schemes, contributing to sustainable agricultural development.

Overall the system aims to bridge the gap between traditional credit assessment methods and agriculture specific risk realities , delivering a scalable and practical solution for rural financial ecosystems.